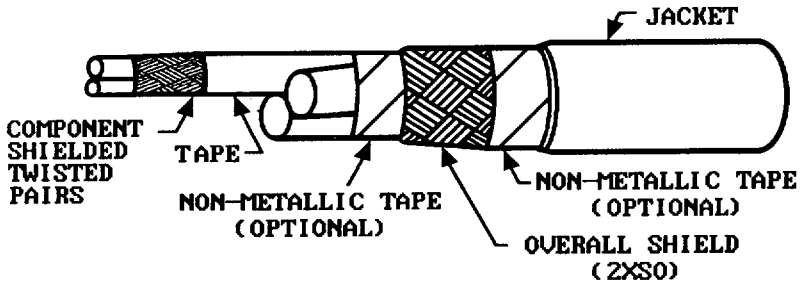
INCH-POUND
MIL-C-24640/9B
31 July 1989
SUPERSEDING
MIL-C-0024640/9A(SH)
23 October 1985
USED IN LIEU OF
MIL-C-24640/9
10 August 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, AWG 22, TYPE 2XS (INCLUDING VARIATION TYPES 2XSA AND 2XSO)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.



SH 12756

COMPONENT SHIELDED PAIR:

(B) One black MIL-W-81044/12-22 wire twisted with one white MIL-W-81044/12-22 wire, except identification codes and methods shall be in accordance with MIL-C-24640. Twisted pair identification code applied by method 2 over shields and method 6 on component wires. The shield shall be tin-coated copper braid with a coverage not less than 90 percent determined in accordance with MIL-C-27500.

FIGURE 1. Construction (non-watertight).

(B) denotes changes.

AMSC N/A
FSC 6145
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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(B) TABLE I. Construction details-unarmored.

PIN M24640/9	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
				(min)	(max)	
-01UN	2XS-2	2	0.038	0.308	0.332	61.4
-02UN	2XS-3	3	.038	.325	.350	74.2
-03UN	2XS-7	7	.041	.423	.455	132.0
-04UN	2XS-10	10	.044	.537	.579	193.0
-05UN	2XS-14	14	.044	.582	.627	240.0
-06UN	2XS-19	19	.044	.644	.694	308.0
-07UN	2XS-24	24	.047	.758	.818	390.0
-08UN	2XS-30	30	.047	.804	.866	468.0

(B) TABLE II. Construction details - armored.

PIN M24640/9	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
				(min)	(max)	
-01AN	2XSA-2	2	0.038	0.358	0.382	81.4
-02AN	2XSA-3	3	.038	.375	.400	101.7
-03AN	2XSA-7	7	.041	.473	.505	176.1
-04AN	2XSA-10	10	.044	.587	.629	228.5
-05AN	2XSA-14	14	.044	.632	.677	280.0
-06AN	2XSA-19	19	.044	.694	.744	359.5
-07AN	2XSA-24	24	.047	.808	.868	442.8
-08AN	2XSA-30	30	.047	.854	.916	531.3

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(B) TABLE III. Construction details-overall shield.

PIN M24640/9	Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)
-01UO	2XSO-3	3	0.271	0.038	0.380	0.410	117.0
-02UO	2XSO-7	7	.373	.041	.474	.510	184.0
-03UO	2XSO-10	10	.487	.044	.594	.640	262.0
-04UO	2XSO-14	14	.541	.044	.636	.686	301.0
-05UO	2XSO-19	19	.607	.044	.709	.765	383.0
-06UO	2XSO-30	30	.777	.047	.869	.937	554.0

REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

When this cable is manufactured with armor, the following additional requirements shall apply:

(B) (a) Braided metal armor shall be added.
(b) Cable surface marking will not be required.

When this cable is manufactured with an overall shield, the following additional requirements shall apply:

(a) An overall tin-coated copper braid shall be added.
(B) (b) Braid coverage shall be not less than 90 percent determined in accordance with MIL-C-27500.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
(B) Conductor resistance - ohms/1000 feet, at 25°C, maximum	17.37
(B) Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield (type 2XSO only)	500
Insulation resistance - megohms/1000 feet, minimum	5000
Conductor and shield continuity (type 2XSO only)	No failure
Jacket flaws (type 2XSO only)	No failure

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EXAMINATION AND TESTS: - Continued	Requirements
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	50
Capacitance unbalance - percent, maximum	15
(B) Characteristic impedance - at 1 MHz, ohms	40 min, 63 max
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
Weight	No failure
Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	3
Gas flame (1 hour)	Not applicable
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor (type 2XSA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	--
Surface transfer impedance (type 2XSO only)	
Milliohms per meter, maximum	700
EMP response, dB, minimum	60
Pulse response time - ms/500 feet	
Rise time, maximum	--
Decay time, maximum	--

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

	Requirements
Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Finished wire	Not applicable

Preparing activity:
Navy - SH
(Project 6145-1004-09)

Custodians:
Army - MI
Navy - SH
Air Force - 85

Review activities:
Navy - EC
Air Force - 99
DLA - IS

User activities:
Army - ME, AR
Navy - CG

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